1 The SYK model and its partition function

Fix integers $N, q \in \mathbb{N}$ such that $0 < q \leq N$ and q is even. Write

$$\mathcal{I}_{N,q} = \{I \subseteq \{1, \ldots, N\} : |I| = q\}. \tag{1}$$

Thus, an interaction label is a q-element subset of the Majorana labels.

Let $\mathcal{A}$ be a complex star algebra with generators $\psi_1, \ldots, \psi_N \in \mathcal{A}$ satisfying the Majorana relations

$$\psi_i \psi_j + \psi_j \psi_i = 2\delta_{ij}, \qquad \psi_i^* = \psi_i. \tag{2}$$

For $I = \{i_1 < \cdots < i_q\} \in \mathcal{I}_{N,q}$, define the ordered Majorana monomial

$$\psi_I = \psi_{i_1} \cdots \psi_{i_q}. \tag{3}$$

Reversing the order of the product gives

$$\psi_I^* = (-1)^{q/2} \psi_I, \tag{4}$$

where the sign is understood as a scalar in $\mathbb{C}$. Consequently, the phase

$$\phi_q = i^{q/2} \tag{5}$$

produces a self-adjoint Hamiltonian from arbitrary real couplings $J = (J_I)_{I \in \mathcal{I}_{N,q}}$:

$$H(J) = \sum_{I \in \mathcal{I}_{N,q}} J_I \phi_q \psi_I, \qquad H(J)^* = H(J). \tag{6}$$

For the thermodynamic quantities, take a finite-dimensional matrix representation $H \in \mathrm{Mat}_d(\mathbb{C})$ of (6). For inverse temperature $\beta \geq 0$, set

$$\Gamma_\beta(H) = \exp(-\beta H), \tag{7}$$

$$Z(\beta; H) = \mathrm{Tr}\big(\Gamma_\beta(H)\big), \tag{8}$$

$$\langle O \rangle_{\beta,H} = \frac{\mathrm{Tr}\big(O\Gamma_\beta(H)\big)}{Z(\beta; H)}. \tag{9}$$

The trace is unnormalized, so $\Gamma_0(H) = I_d$ and $Z(0; H) = d$.

2 Gaussian disorder and two replicas

To obtain the random SYK model, let $(g_I)_{I\in\mathcal{I}_{N,q}}$ be independent standard real Gaussian variables and choose a deterministic scale $\sigma_N \geq 0$. We use the couplings

$$J_I = \sigma_N g_I, \qquad Z_N(\beta; g) = Z\big(\beta; H(J)\big), \qquad F_N(\beta) = \mathbb{E}\big[Z_N(\beta; g)\big]. \tag{10}$$

Let $\widetilde{g}$ be an independent copy of g. For $\rho \in [-1, 1]$, define the correlated disorder

$$g_I^{(\rho)} = \rho g_I + \sqrt{1-\rho^2}\,\widetilde{g}_I, \qquad I \in \mathcal{I}_{N,q}, \tag{11}$$

and the two-replica partition function

$$\mathcal{G}_N(\rho, \beta) = \mathbb{E}\left[Z_N(\beta; g) Z_N\big(\beta; g^{(\rho)}\big)\right]. \tag{12}$$

Proposition 1 (Mehler decomposition)**.** *Let P_r denote orthogonal projection onto the rth Gaussian Wiener chaos. Then*

$$\mathcal{G}_N(\rho, \beta) = F_N(\beta)^2 + \sum_{r\geq 1} \rho^r \left\| P_r Z_N(\beta; g)\right\|_{L^2}^2. \tag{13}$$

In particular,

$$\mathbb{E}\left[Z_N(\beta; g)^2\right] = F_N(\beta)^2 + \sum_{r\geq 1} \left\| P_r Z_N(\beta; g)\right\|_{L^2}^2. \tag{14}$$

3 A correlated two-replica calculation for the partition function

Throughout this section, let

$$F_N(\beta) = \mathbb{E} Z_N(\beta). \tag{15}$$

Let $\widetilde{g} = (\widetilde{g}_I)_{I\in\mathcal{I}_N}$ be an independent copy of $g = (g_I)_{I\in\mathcal{I}_N}$. For $\rho \in [-1, 1]$, define

$$g_I^{(\rho)} = \rho g_I + \sqrt{1-\rho^2}\,\widetilde{g}_I, \qquad I \in \mathcal{I}_N, \tag{16}$$

and set

$$\mathcal{G}_N(\rho, \beta) = \mathbb{E}\left[Z_N(\beta; g) Z_N(\beta; g^{(\rho)})\right]. \tag{17}$$

Proposition 2 (Mehler decomposition)**.** *For every $\rho \in [-1,1]$ and $\beta \in \mathbb{R}$,*

$$\mathcal{G}_N(\rho,\beta) = F_N(\beta)^2 + \sum_{r\geq 1} \rho^r \|P_r Z_N(\beta)\|_2^2. \tag{18}$$

In particular,

$$\mathbb{E}[Z_N(\beta)^2] = F_N(\beta)^2 + \sum_{r\geq 1} \|P_r Z_N(\beta)\|_2^2. \tag{19}$$

Proof. Write the Hermite expansion as

$$Z_N(\beta; g) = \sum_{\alpha} z_{N,\alpha}(\beta) H_\alpha(g). \tag{20}$$

The correlated Gaussian identity gives

$$\mathbb{E}\left[H_\alpha(g) H_\gamma(g^{(\rho)})\right] = \mathbf{1}_{\{\alpha=\gamma\}} \rho^{|\alpha|} \alpha!. \tag{21}$$

Substituting (20) into (17) and grouping terms according to $|\alpha|$ proves (18). Taking $\rho = 1$ proves (19). □

Lemma 3 (Exact first and second chaoses of Z_N)**.** *The first chaos vanishes:*

$$P_1 Z_N(\beta) = 0. \tag{22}$$

Moreover, there exists a scalar $b_N(\beta)$ such that

$$P_2 Z_N(\beta) = b_N(\beta) \sum_{I\in\mathcal{I}_N} (g_I^2 - 1), \tag{23}$$

and

$$2M_N b_N(\beta) = \beta F_N'(\beta). \tag{24}$$

Consequently,

$$\|P_2 Z_N(\beta)\|_2^2 = \frac{\beta^2}{2M_N} F_N'(\beta)^2. \tag{25}$$

Proof. The same Majorana sign-gauge transformation used for L_N gives

$$Z_N(\beta, g) = Z_N(\beta, \chi_s g), \qquad s \in \{\pm 1\}^N. \tag{26}$$

The corresponding Hermite parity rule implies (22) and eliminates all off-diagonal terms $g_I g_J$ with $I \neq J$ from the second chaos. Permutation symmetry then gives (23).

By the Hermite coefficient formula,

$$b_N(\beta) = \frac{1}{2}\mathbb{E}[\partial_{g_I}^2 Z_N(\beta)]. \tag{27}$$

Summing over I and applying Gaussian integration by parts yields

$$\begin{aligned} 2M_N b_N(\beta) &= \sum_{I\in\mathcal{I}_N} \mathbb{E}[\partial_{g_I}^2 Z_N(\beta)] \\ &= \sum_{I\in\mathcal{I}_N} \mathbb{E}[g_I \partial_{g_I} Z_N(\beta)]. \end{aligned} \tag{28}$$

Since $H_N(g)$ is linear in g,

$$\sum_{I\in\mathcal{I}_N} g_I \partial_{g_I} Z_N(\beta) = \beta \partial_\beta Z_N(\beta). \tag{29}$$

Taking expectation proves (24). Finally, $\mathrm{Var}(g_I^2 - 1) = 2$, so (25) follows from (23) and (24). □

Corollary 4 (Exact second-moment reduction)**.** *For every $\beta \in \mathbb{R}$,*

$$\frac{\mathbb{E}[Z_N(\beta)^2]}{F_N(\beta)^2} = 1 + \frac{\beta^2}{2M_N}\left(\partial_\beta \log F_N(\beta)\right)^2 + \mathcal{R}^Z_{N,\geq 3}(\beta), \tag{30}$$

where

$$\mathcal{R}^Z_{N,\geq 3}(\beta) = \frac{1}{F_N(\beta)^2}\sum_{r\geq 3}\|P_r Z_N(\beta)\|_2^2 \geq 0. \tag{31}$$

More generally,

$$\frac{\mathcal{G}_N(\rho,\beta)}{F_N(\beta)^2} = 1 + \rho^2\frac{\beta^2}{2M_N}\left(\partial_\beta \log F_N(\beta)\right)^2 + \sum_{r\geq 3}\rho^r \frac{\|P_r Z_N(\beta)\|_2^2}{F_N(\beta)^2}. \tag{32}$$

Proof. Combine Proposition 2 with Lemma 3. □

We next compute the first two non-zero Taylor coefficients of (30). Define

$$S_N = \sum_{I,J\in\mathcal{I}_N}(-1)^{|I\cap J|} = M_N \sum_{j=0}^{q}(-1)^j \binom{q}{j}\binom{N-q}{q-j}, \tag{33}$$

and

$$U_N = \begin{cases} \dfrac{N!}{6\left(N-\frac{3q}{2}\right)!\left(\left(\frac{q}{2}\right)!\right)^3}, & N \geq \frac{3q}{2}, \\ 0, & N < \frac{3q}{2}. \end{cases} \tag{34}$$

Proposition 5 (Taylor expansion of the normalized second moment)**.** *For fixed N and $\beta \to 0$,*

$$F_N(\beta) = 1 + \frac{\beta^2}{2}\sigma_N^2 M_N + \frac{\beta^4}{24}\sigma_N^4 \left(2M_N^2 + S_N\right) + O_{N,q}(\beta^6). \tag{35}$$

Moreover,

$$\begin{aligned}\frac{\mathbb{E}[Z_N(\beta)^2]}{F_N(\beta)^2} &= 1 + \frac{\beta^4}{2}\sigma_N^4 M_N \\ &\quad + \beta^6\sigma_N^6 \left\{\frac{S_N - M_N^2}{6} + \mathbf{1}_{\{4|q\}} U_N\right\} + O_{N,q}(\beta^8).\end{aligned} \tag{36}$$

Proof. The Clifford relations imply

$$\operatorname{tr}(\psi_I\psi_J) = \mathbf{1}_{\{I=J\}}, \qquad \operatorname{tr}(\psi_I\psi_J\psi_I\psi_J) = (-1)^{|I\cap J|}. \tag{37}$$

Hence Wick's formula gives

$$\mathbb{E}\operatorname{tr}(H_N^2) = \sigma_N^2 M_N, \qquad \mathbb{E}\operatorname{tr}(H_N^4) = \sigma_N^4\left(2M_N^2 + S_N\right). \tag{38}$$

Expanding the exponential proves (35). It follows that

$$\partial_\beta \log F_N(\beta) = \sigma_N^2 M_N \beta + \frac{\sigma_N^4}{6}\left(S_N - M_N^2\right)\beta^3 + O_{N,q}(\beta^5). \tag{39}$$

Therefore, the second-chaos term in (30) equals

$$\begin{aligned}\frac{\beta^2}{2M_N}\left(\partial_\beta \log F_N(\beta)\right)^2 &= \frac{\beta^4}{2}\sigma_N^4 M_N \\ &\quad + \frac{\beta^6}{6}\sigma_N^6\left(S_N - M_N^2\right) + O_{N,q}(\beta^8).\end{aligned} \tag{40}$$

It remains to identify the order-β^6 contribution of the third chaos. Every third-chaos multi-index with a repeated hyperedge label violates the Majorana gauge-parity rule. Thus only products $g_I g_J g_K$ with distinct $I, J, K \in \mathcal{I}_N$ can contribute. The coefficient of such a product in the order-β^3 term of $Z_N(\beta)$ can be non-zero only if

$$I \triangle J \triangle K = \varnothing. \tag{41}$$

In that case, every vertex belongs to exactly two of I, J, K, and hence

$$|I \cap J| = |I \cap K| = |J \cap K| = \frac{q}{2}. \tag{42}$$

The number of unordered triples satisfying (41) is U_N. The six orderings of such a triple have the same trace when $4 \mid q$, whereas they cancel in pairs when $q \equiv 2 \pmod 4$. Since

$$|\mathrm{tr}(\psi_I \psi_J \psi_K)| = 1, \tag{43}$$

we obtain

$$\|P_3 Z_N(\beta)\|_2^2 = \mathbf{1}_{\{4|q\}} \beta^6 \sigma_N^6 U_N + O_{N,q}(\beta^8). \tag{44}$$

The total contribution of chaoses of order at least four is $O_{N,q}(\beta^8)$. Substituting (40) and (44) into (30) proves (36). □

Corollary 6 (Large-N asymptotics of the first Taylor coefficients)**.** *As $N \to \infty$,*

$$\frac{S_N}{M_N^2} = 1 - \frac{2q^2}{N} + O_q(N^{-2}), \tag{45}$$

and

$$\frac{1}{2}\sigma_N^4 M_N = \frac{(q-1)!}{2q} N^{2-q} \left(1 + O_q(N^{-1})\right). \tag{46}$$

Furthermore,

$$\frac{1}{6}\sigma_N^6 \left(S_N - M_N^2\right) = -\frac{(q-1)!}{3} N^{2-q} \left(1 + O_q(N^{-1})\right), \tag{47}$$

whereas

$$\sigma_N^6 U_N = O_q\left(N^{3-3q/2}\right) = o\left(N^{2-q}\right). \tag{48}$$

Consequently,

$$\lim_{N\to\infty} N^{q-2} \frac{1}{4!}\, \partial_\beta^4 \frac{\mathbb{E}[Z_N(\beta)^2]}{F_N(\beta)^2}\bigg|_{\beta=0} = \frac{(q-1)!}{2q}, \tag{49}$$

and

$$\lim_{N\to\infty} N^{q-2} \frac{1}{6!}\, \partial_\beta^6 \frac{\mathbb{E}[Z_N(\beta)^2]}{F_N(\beta)^2}\bigg|_{\beta=0} = -\frac{(q-1)!}{3}. \tag{50}$$

Proof. For fixed $I \in \mathcal{I}_N$ and uniformly sampled $J \in \mathcal{I}_N$, $|I \cap J|$ has the hypergeometric distribution. Therefore,

$$\frac{S_N}{M_N^2} = \mathbb{E}\left[(-1)^{|I\cap J|}\right] = 1 - 2\mathbb{P}(|I \cap J| \text{ is odd}). \tag{51}$$

Since

$$\mathbb{P}(|I \cap J| = 1) = \frac{q\binom{N-q}{q-1}}{\binom{N}{q}} = \frac{q^2}{N} + O_q(N^{-2}), \tag{52}$$

and $\mathbb{P}(|I \cap J| \geq 2) = O_q(N^{-2})$, (45) follows. The remaining statements follow from $M_N \sim N^q/q!$, $\sigma_N^2 = (q-1)!N^{-(q-1)}$, and $U_N = O_q(N^{3q/2})$. □

Remaining high-temperature estimate. Corollary 4 shows that a CLT for $Z_N(\beta)$ follows once one proves, for fixed sufficiently small $\beta \neq 0$,

$$\frac{\mathbb{E}[Z_N(\beta)^2]}{F_N(\beta)^2} = 1 + \frac{\beta^2}{2M_N} \left(\partial_\beta \log F_N(\beta)\right)^2 + o\left(N^{2-q}\right). \tag{53}$$

Indeed, all terms in (31) are non-negative, so (53) is equivalent to

$$\sum_{r\geq 3} \|P_r Z_N(\beta)\|_2^2 = o\left(F_N(\beta)^2 N^{2-q}\right). \tag{54}$$

The Taylor calculation above verifies the expected second-chaos dominance through order β^6: the third-chaos contribution at that order is smaller by a factor of order $N^{1-q/2}$.

References

[GPSZ26] David Gamarnik, Francisco Pernice, Alexander Schmidhuber, and Alexander Zlokapa. The free energy limit of the SYK model at high temperature. arXiv:2605.02768, 2026.

[ACKK25] Eric R. Anschuetz, Chi-Fang Chen, Bobak T. Kiani, and Robbie King. Strongly interacting fermions are non-trivial yet non-glassy. Physical Review Letters, 135(3):030602, 2025.

[Wil67] R. M. Wilcox. Exponential operators and parameter differentiation in quantum physics. Journal of Mathematical Physics, 8(4):962–982, 1967.